

pi0.5 — Outlier magnitude heatmap ($\log_{10}(|x|/\text{scale})$) under per-block MSE-optimal int4 scaling)
Blue = crushed to int4 level 0; white ≈ 1 int4 level; red = saturating ceiling ($\log_{10}(7.5) \approx 0.87$)

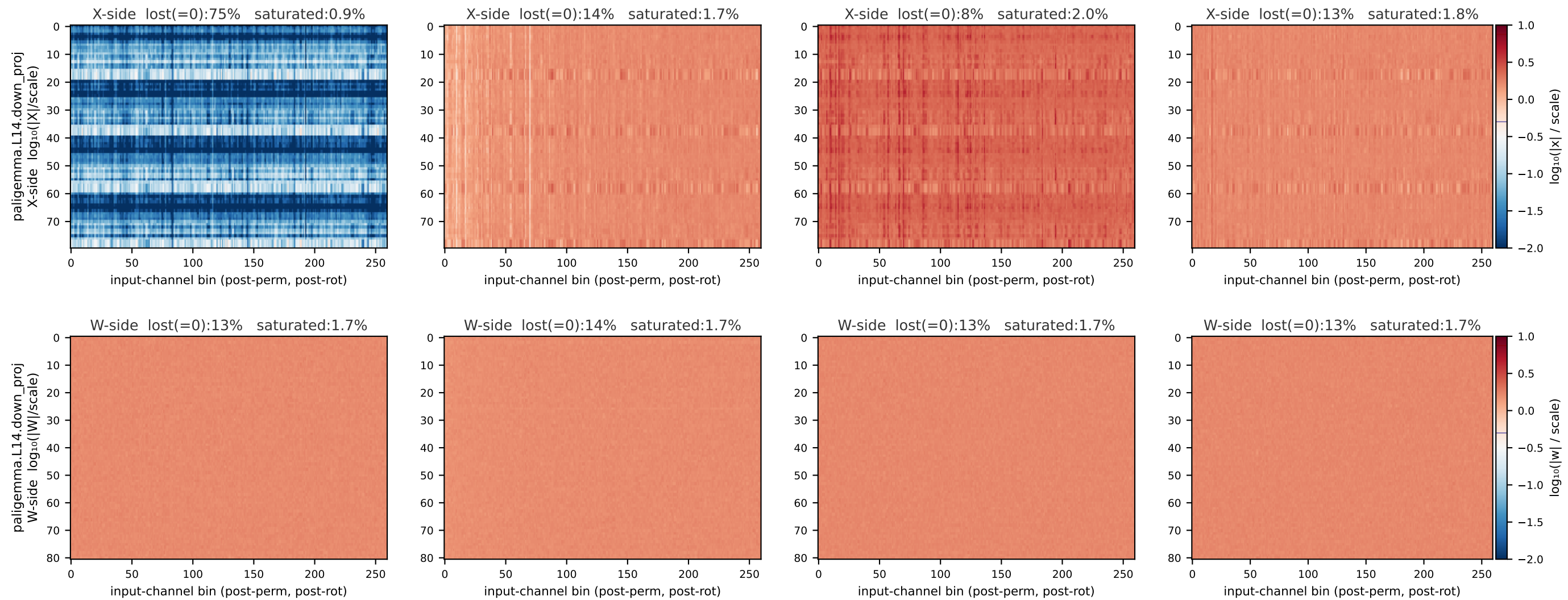
raw (no rotation)

DuQuant SVD

Hadamard only

SVD-Hadamard (A2-lite)

paligemma.L14.down_proj — pathology (X channel-skew 156 $\times$)



paligemma.L02.gate_proj — typical MLP layer (SVD-only stuck at 12.6 $\times$)

